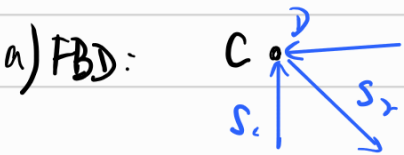
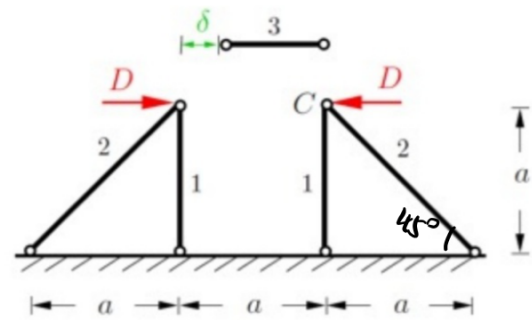


Homework 1.7 Truss member 3 was produced too short to be assembled between two identical trusses.

a) Determine the required assembly force D ?

b) Calculate the normal force S_3 after the assembly ($D = 0$)?

Given: $EA_1 = EA_3 = EA$, $EA_2 = \sqrt{2}EA$.



$$+\vec{x}: D - S_2 \cos 45^\circ = 0$$

$$+\vec{y}: S_1 - S_2 \sin 45^\circ = 0$$

$$D = \frac{\sqrt{2}}{2} S_2 = S_1$$

$$V = \frac{\Delta l_2}{\sin 45^\circ} + \frac{u}{\tan 45^\circ}$$

$$V = \sqrt{2} \Delta l_2 + \Delta l_1 = \frac{\delta}{2}$$

$$2 \sqrt{2} \left(\frac{\sqrt{2} S_1 a}{EA} \right) + \frac{S_1 a}{EA} = \frac{\delta}{2}$$

$$\Delta l_1 = \frac{S_1 a}{EA}$$

$$\Delta l_2 = \frac{S_2 \sqrt{2} a}{\sqrt{2} EA} = \frac{S_2 a}{EA} = \frac{\sqrt{2} S_1 a}{EA}$$

$$\frac{3 S_1 a}{EA} = \frac{\delta}{2}$$

$$S_1 = \frac{EA \delta}{6a} = D$$

$$V^* + \Delta l_3 = \delta \quad \Delta l_3 = \frac{S_3(a-\delta)}{EA}$$

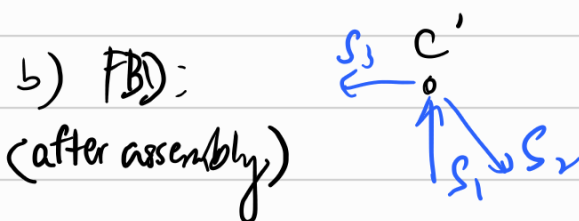
$$\frac{6 S_3 a}{EA} + \frac{S_3(a-\delta)}{EA} = \delta$$

$$6 S_3 a + S_3(a-\delta) = EA \delta$$

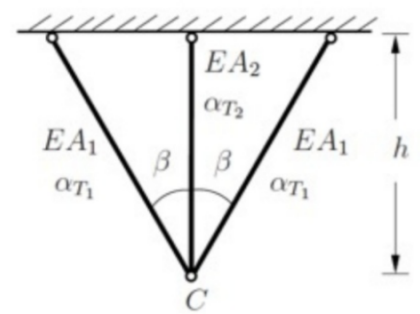
$$7 S_3 a - S_3 \delta = EA \delta$$

$$S_3 (7a - \delta) = EA \delta$$

$$S_3 = \frac{\delta}{7a - \delta} (EA)$$

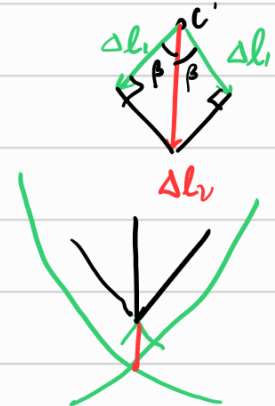
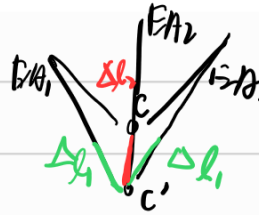


Homework 1.8 In the depicted truss the members have the axial rigidities EA_1 , EA_2 and the coefficients of thermal expansion α_{T1} , α_{T2} . Determine the axial forces in the trusses, if the system is heated by ΔT ?



$$2S_1 \cos \beta + S_2 = 0$$

$$S_2 = -2S_1 \cos \beta \quad \cos \beta = \frac{\Delta l_1}{\Delta l_2} \quad \Delta l_1 = \Delta l_2 \cos \beta$$



By Hooke's law for thermal elasticity,

$$\Delta l = \frac{Fl}{EA} + \alpha_T \Delta T l$$

$$\Delta l_1 = \frac{S_1 l_1}{EA_1} + \alpha_{T1} \Delta T l_1$$

$$\Delta l_2 = \frac{S_2 l_2}{EA_2} + \alpha_{T2} \Delta T l_2$$

$$\frac{S_1 l_1}{EA_1} + \alpha_{T1} \Delta T l_1 = \left(\frac{-2S_1 \cos \beta l_2}{EA_2} + \alpha_{T2} \Delta T l_2 \right) \cos \beta$$

$$\frac{S_1 h}{EA_1 \cos \beta} + \alpha_{T1} \Delta T \frac{h}{\cos \beta} = \frac{-2S_1 \cos^2 \beta h}{EA_2} + \alpha_{T2} \Delta T h \cos \beta$$

$$l_2 = h \quad l_1 = \frac{h}{\cos \beta}$$

$$\frac{S_1 h}{EA_1 \cos \beta} + \frac{2S_1 \cos^2 \beta h}{EA_2} = \Delta T h (\alpha_{T2} \cos \beta - \frac{\alpha_{T1}}{\cos \beta})$$

$$S_2 = -2S_1 \cos \beta$$

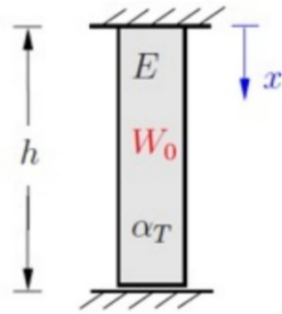
$$= -2 \left(\frac{\alpha_{T2} \cos^2 \beta - \alpha_{T1} \cos \beta}{1 + 2 \cos^3 \beta \frac{EA_1}{EA_2}} \right) \Delta T EA_1$$

$$S_1 \frac{h}{\cos \beta} \left(\frac{1}{EA_1} + \frac{2 \cos^2 \beta}{EA_2} \right) = \Delta T h \left(\frac{\alpha_{T2} \cos^2 \beta - \alpha_{T1}}{\cos \beta} \right)$$

$$S_1 = \left(\frac{\alpha_{T2} \cos^2 \beta - \alpha_{T1}}{1 + 2 \cos^3 \beta \frac{EA_1}{EA_2}} \right) \Delta T EA_1$$

Homework 1.9 A massive bar (weight W_0 , cross section area A , thermal expansion coefficient α_T) is fixed at $x = 0$ and just touches the ground in a stress-free manner.

Determine the stress distribution $\sigma(x)$ in the bar after a uniform heating by ΔT . Which ΔT causes compression everywhere in the bar?



$$\sigma_w = \frac{N}{A} = \frac{W(x)}{A} = \frac{W_0(1 - \frac{x}{h})}{A} \quad w(x) = W_0(1 - \frac{x}{h})$$

$$\epsilon_T = \frac{\sigma}{E} + \alpha_T \Delta T \quad \text{since bar is in contact with ground, } \therefore \text{elongation} = 0$$

$$0 = \frac{\sigma}{E} + \alpha_T \Delta T \quad \sigma = \sigma_i + \sigma_w = \frac{W_0(1 - \frac{x}{h})}{A} - E \alpha_T \Delta T$$

$$\sigma_T = -E \alpha_T \Delta T$$

Compressive stress everywhere in the bar as long as compression is present at both ends of bar.

There will always be compressive stress at the bottom of bar due to thermal expansion from uniform heating.

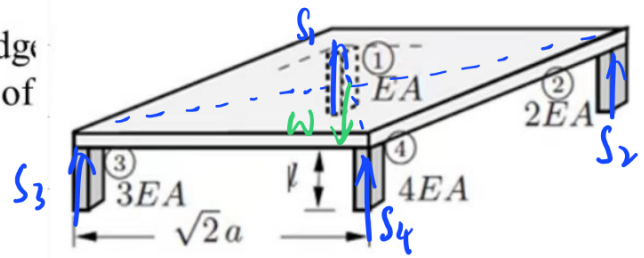
$$\text{As for top } (x=0), \sigma < 0 \quad \frac{W_0(1 - \frac{0}{h})}{A} - E \alpha_T \Delta T < 0$$

$$\Delta T > \frac{W_0}{E A \alpha_T}$$

Homework 1.10 A rigid quadratic plate (weight W , edge length $\sqrt{2} a$) is supported on 4 elastic posts. The posts are of equal length l , but possess different axial rigidities.

Determine

- the weight distribution on the 4 posts;
- the displacement f in the middle of the plate.



a) $S_1 + S_2 + S_3 + S_4 = W$

$\sum M_O = aS_4 - aS_1 = 0$
 $S_4 = S_1$

$\sum M_O = aS_3 - aS_2 = 0$
 $S_3 = S_2$

$f = \frac{1}{2}(v_1 + v_4) = \frac{1}{2}(v_2 + v_3)$
 $v_i = \frac{F_i l}{EA}$

$2S_1 + 2S_2 = W$

$2(S_1 + S_2) = W$

$2\left(\frac{2}{3}S_2 + S_2\right) = W$

$2\left(\frac{5}{3}\right)S_2 = W$

$\frac{10}{3}S_2 = W$

$S_2 = \frac{3}{10}W$

$f = \frac{1}{2}\left(\frac{S_1 l}{EA} + \frac{S_4 l}{4EA}\right) = \frac{1}{2}\left(\frac{S_2 l}{2EA} + \frac{S_3 l}{3EA}\right)$

$\frac{5}{4}S_1 l = \frac{5}{6}S_2 l$

$S_1 l = \frac{2}{3}S_2 l$

$S_1 = \frac{2}{3}\left(\frac{3}{10}\right)W$
 $= \frac{1}{5}W$

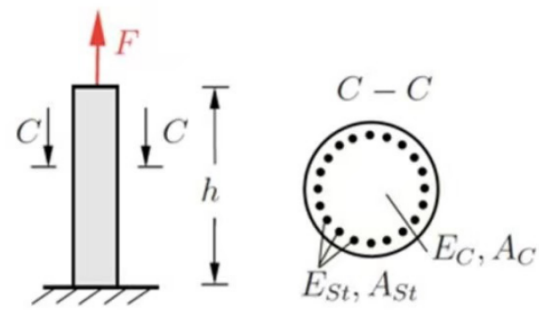
$f = \frac{1}{2}\left(\frac{\frac{1}{5}Wl}{EA} + \frac{\frac{1}{5}Wl}{4EA}\right) = \frac{1}{8}\frac{Wl}{EA}$

Homework 1.11 A column of steel reinforced concrete is loaded by a tensile force F . What are the stresses in the concrete and the steel as well as the height change Δh of the column, if we assume

a) a perfect bonding between steel and concrete?

b) the concrete is cracked and does not carry any load?

Given: $E_{St}/E_C = 6$, $A_{St}/A_C = 1/9$.



FBD:



a)

$$N_c + N_{st} = F$$

$$\Delta h = \Delta h_{st} = \Delta h_c$$

$$\Delta h_{st} = \frac{N_{st} h}{E_{st} A_{st}} \quad \Delta h_c = \frac{N_c h}{E_c A_c}$$

$$\frac{E_{st} A_{st}}{E_c A_c} = \frac{6}{1} = \frac{2}{3}$$

$$E_{st} A_{st} = \frac{2}{3} E_c A_c$$

$$\frac{3}{2} E_{st} A_{st} = E_c A_c$$

$$N_{st} = \frac{\Delta h_{st} E_{st} A_{st}}{h} \quad N_c = \frac{\Delta h_c E_c A_c}{h}$$

$$\frac{\Delta h}{h} (E_{st} A_{st} + E_c A_c) = F$$

$$\frac{5}{2} \left(\frac{\Delta h}{h} E_{st} A_{st} \right) = F$$

$$N_{st} = \frac{2}{5} F \quad N_c = \frac{3}{5} F$$

$$\Delta h = \frac{\frac{2}{5} F h}{E_{st} A_{st}}$$

$$A = A_{st} + A_c$$

$$A_{st} = \frac{1}{9} A_c$$

$$A = \frac{10}{9} A_c$$

$$A_c = \frac{9}{10} A \quad A_{st} = \frac{1}{10} A$$

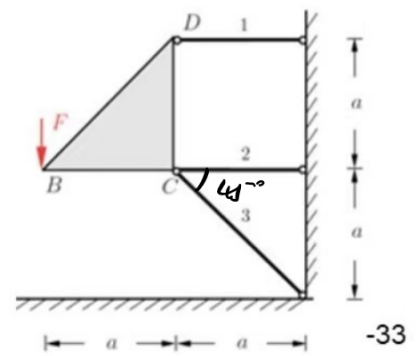
$$\sigma_{st} = \frac{\frac{2}{5} F}{\frac{1}{10} A} = 4 \frac{F}{A} \quad \sigma_c = \frac{\frac{3}{5} F}{\frac{9}{10} A} = \frac{2}{3} \frac{F}{A}$$

b) if only steel carries load,

$$\sigma = \frac{F}{A_{st}} = \frac{F}{\frac{1}{10} A} = 10 \frac{F}{A} \quad \Delta h = \frac{F h}{E_{st} A_{st}}$$

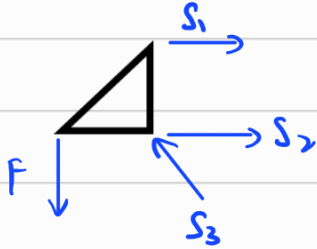
Homework 1.12 A rigid weightless triangle is supported by 3 bars with the axial rigidity EA . The triangle is loaded in point B by the force F . Determine

- the forces S_i in the 3 bars and their elongations Δl_i ;
- the displacement of point C.



-33

a) FBD:



$$+\uparrow: \frac{\sqrt{2}}{2} S_3 - F = 0 \quad +\curvearrowleft: aF - aS_1 = 0 \quad S_1 = F$$

$$S_3 = \sqrt{2}F$$

$$+\vec{n} = \frac{\sqrt{2}}{2} S_3 - S_1 - S_2 = 0$$

$$\sqrt{2}F - F - S_2 = 0$$

$$S_2 = (\sqrt{2} - 1)F$$

Check this!

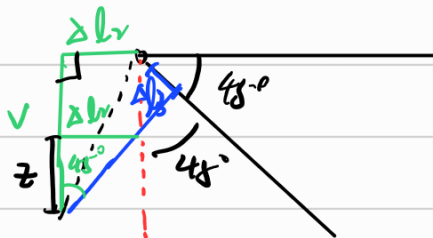
$$\Delta l_1 = \frac{S_1 a}{EA} = \frac{Fa}{EA}$$

$$\Delta l_2 = \frac{S_2 a}{EA} = \frac{(\sqrt{2} - 1)Fa}{EA}$$

$$\Delta l_3 = \frac{S_3 \frac{a}{\sqrt{2}}}{EA} = \frac{2Fa}{EA}$$

Wrong answer

b)



$$v = \frac{\Delta l_2}{\tan 45^\circ} + \frac{\Delta l_3}{\sin 45^\circ}$$

$$= \Delta l_2 + \sqrt{2} \Delta l_3$$

$$= \frac{(\sqrt{2} - 1)Fa}{EA} + \frac{2\sqrt{2}Fa}{EA}$$

$$= \frac{(3\sqrt{2} - 1)Fa}{EA}$$

Wrong answer.